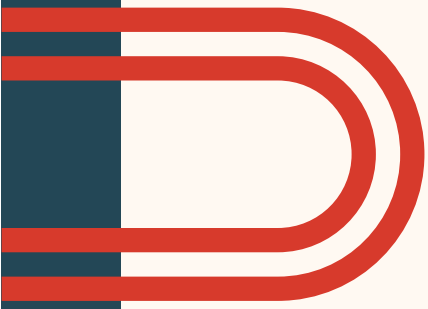




ASPire

EC STUDENT PROBATION DATABASE

GROUP 2

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OVERVIEW



A database built to facilitate and improve the probation recovery program for ESC students. The database will aid in assessing information, improving student experience in the college of ESC, and assist advisors in maintaining accurate and timely records.

ECS may place any student who has a cumulative GPA of less than 2.0 and for whom a more serious action is not appropriate on probation. A student may be placed on probation if their semester GPA falls below 2.0, or if the student fails to meet other criteria for good academic standing as established by the school/college.

INTRODUCTION

Student Success Advisors meet regularly with students assigned to them in order to identify barriers to success, find and discuss strategies for improving academic performance, and offer resources and encouragement to stay on track with their plan. A semester framework has been established by the school to provide a base for student plan experience, however meetings are generally student-driven and shape materials to best assist each individual.

This project consists of the creation of a database for students currently on probation. In order to assist students, advisors, and other support staff accurately it will provide insight into the following questions:

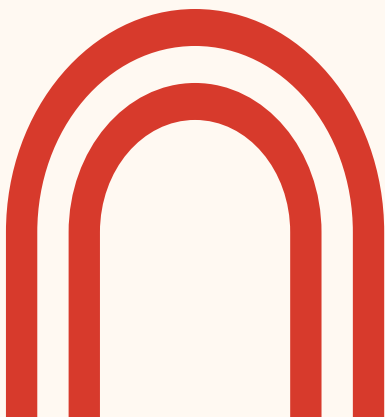
- *How often do ECS students fall into probation?*
- *Is there a seasonal trend (fall vs spring)?*
- *Is there a yearly trend (1st,2nd, 3rd,4th)?*
 - *Is one year more at risk than another?*
- *Average graduation rate for students who fall into probation?*
- *What is the average GPA for students on probation?*
 - *Mean GPA, Range of GPA, etc.*
- *How often do students fall into probation over a 5-year period?*
- *What is the rate of probation in ESC vs. other colleges' courses?*
 - *Do specific courses increase the risk of falling into probation?*
- *How often are students suspended?*
- *What are the student profiles of those in probation?*
- *Do different levels of probation influence the risk of not graduating?*
 - *What % of students are on probation by level type?*

Using a combination of FAFSA data, self-reported surveys, and the central ORACLE database to compile student information, this project will identify trends in data, create a tool for enhanced analytics, and enable students and staff to examine outcomes.

OBJECTIVES

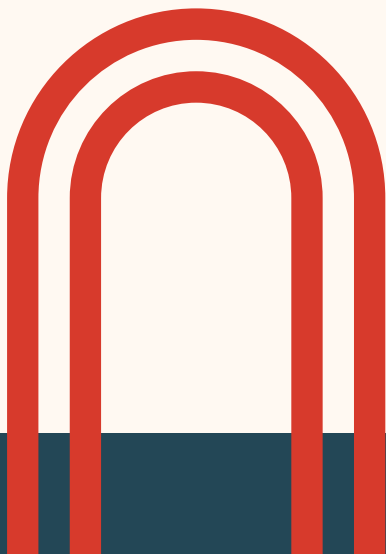
The following objectives shape the rules and operational guidelines of the database. The team will:

- Understand the factors that place students on probation and prevent students from going on probation.
- Increase student retention (which is more cost-effective than new student recruitment).
- Gain insight into the probationary student profile.
- Align student outcomes with curriculum/school needs and goals.
- Reduce number of students on probation (avg 10% of school)

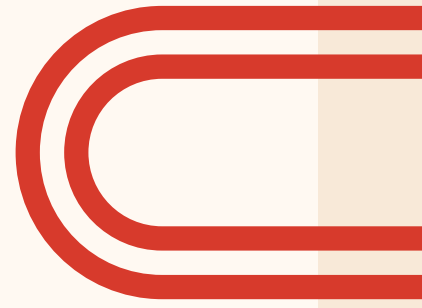


GLOSSARY

<i>Term</i>	<i>Definition</i>
ECS	Engineering & Computer Science
COP	College Probation
PST	Probation: One Semester Trial
PAW	Probation with Advice to Withdraw
IRE-A	Academic Suspension with Appeal – Ineligible to Register Until Appeal Decision
IRE	Academic Suspension – Ineligible to Register



BUSINESS NEEDS



- Success Advisors need to be able to track student probation history and progress
 - o The probation database will allow advisors to:
 - Assess student information and study student profiles
 - Enhance student experiences in ECS
 - Improve the “Thriving in ECS” probation recovery program
- Advisors will be able to identify trends:
 - o Advisors will be able to access meeting dates and notes by student
 - o Advisors will be able to compare probation enrollment by date, season, semester, etc.
- Advisors will be able to perform enhanced analytics:
 - o Advisors will be able to view academic history by student department and major
 - o Advisors can examine graduation rates, GPA data, and frequency of suspensions
- Advisors will be able to predict and shape outcomes:
 - o Advisors will be able to enter appropriate probation notes for students
 - o Advisors can reduce the number of students on probation
 - o Advisors can increase student retention and implement university and ECS policy to best support students
- Advisors will be able to pull and edit relevant student information pertaining to advising meetings, student progress, and university/department standards.
 - o Detailed student probation histories will allow advisors to best assist individual needs.

CONCLUSION

This project was a larger undertaking than our group first imagined. With extensive planning phases, several edits, and intense collaboration we were able to create a project out of the initial phase of the plan. The group took a multi-pronged approach, both coding as a group and taking individual responsibility for various final deliverables. Along with the efforts to produce the project as it stands, we have taken time to lay some groundwork for potential future changes and growth.

TAKEAWAY #1

Working with student data for this project required careful planning to maintain student privacy while building a functional database. With such a large-scale plan, we needed to take time to anonymize records that will have to be removed and reinserted in later phases if student success advisors want to take advantage of the interface. Changes such as displaying a student ID rather than a student name from the ASPIRE information helped the database run more smoothly but will need to be accounted for later if more changes are made.

TAKEAWAY #2

Part of our project was made easier by having a previously existing database to pull data from and insert into our model. Working with a clear, real-life business need from the start shaped our plans from the early flow diagram through the final logical model. Framing the project functionally and continually asking for feedback from potential users guided the current final result as well as future phases in planning.

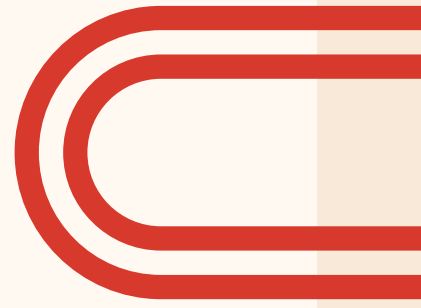
TAKEAWAY #3

This project was far too large to take on all at once. An important realization the team had early on in the process was that it would need to be broken down into phases. The next steps for this project involve at least two more phases before it can be fully functional (as seen in the flow diagram and other models) and more conferencing with users would be needed as well to continually update the design. Goals for the future of the project include adding content provided by student surveys and including another lookup table for other college IDs to expand the program to other colleges.

APPENDIX A

Zip File Contents

Conceptual data model diagram
Logical data model diagram
Identification of the external data model and data logic
Basic application screen layout
Diagram of application screens
SQL up/down script to implement internal model with data
SQL up/down script to load/migrate existing data
SQL up/down script of data logic for external data model
PowerPoint Presentation
Technical report detailing project, data analysis, and team worklog

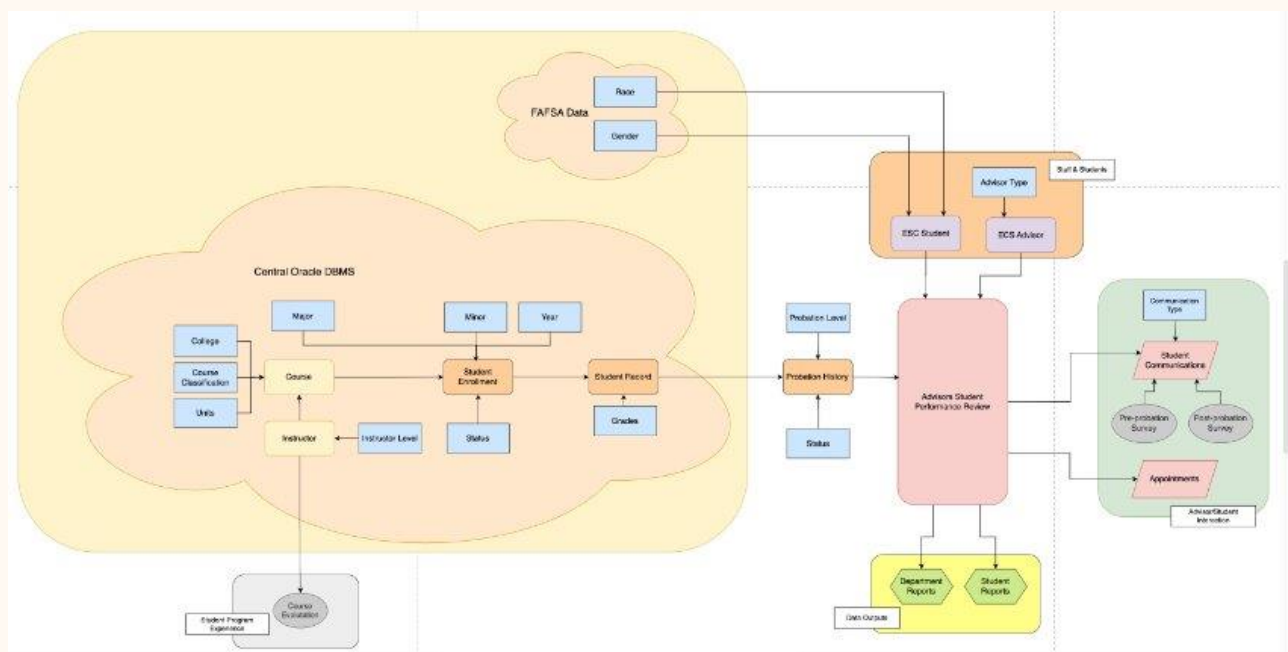


APPENDIX B

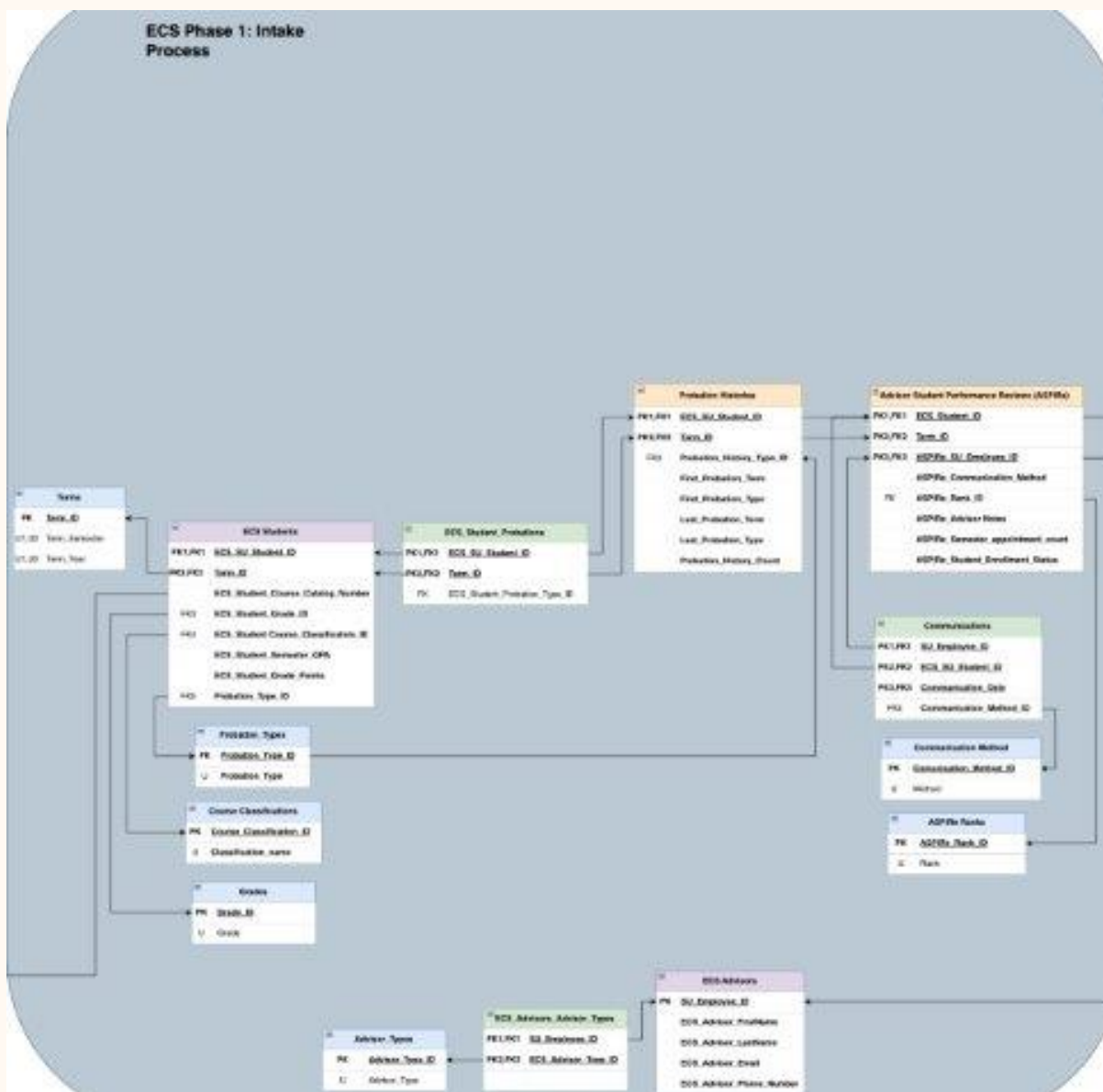
Data Tables – Reference

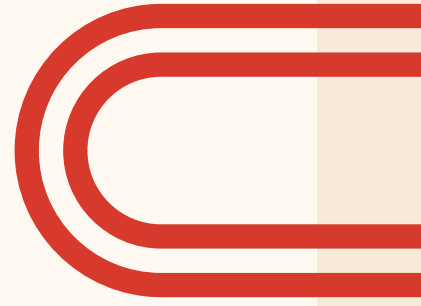
Full-size available in zip file

Flow Diagram:









APPENDIX C

Worklog



Probationary Student Database Worklog

[illegible]

#####	Group Meeting			✓							✓			✓			✓	✓	Done
#####	Group Meeting			✓			✓			✓			✓				✓	✓	Done
#####	Up/Down V3			✓													✓		
#####	Up/Down V4			✓													✓		
#####	After Class Group Meeting			✓			✓			✓			✓				✓	✓	Done
#####	Up/Down V5			✓													✓		
#####	Logical Model V11																✓		
#####	Up/Down V5			✓													✓		
#####	Logical Model V12																✓		
#####	Up/Down V6			✓													✓		
#####	Technical Report V4									✓									
#####	Group Meeting			✓			✓			✓			✓				✓	✓	Done
#####	Group Meeting			✓			✓			✓			✓				✓	✓	Done
#####	Technical Report V5									✓									Done
#####	Final Presentation Live	✓				✓				✓				✓			✓		

SUBMISSION DETAILS

Required artifacts and/or attributes for team project submission are as follows:

✓ Done
✓ Done
✓ Done
✓ Done
✓ Done
✓ Done
✓ Done
✓ Done
✓ Done
✓ Done
✓ Done
✓ Done
✓ Done
✓ Done
✓ Done

1. Technical Report with team name, group members, and which project you will work on. Use APA format to cite any sources used in report. Report should include an introduction/high-level overview, objectives, glossary, specific business rules, and conclusion.
2. Data analysis of the facts listing entities, attributes, and relationships in the data model.
3. Conceptual data model diagram.
4. Logical data model diagram.
5. Identification of your external data model and data logic.
6. Basic layout of all application screens.
7. Diagram of each screen used in the application.
8. SQL up/down script to implement the internal model with initial data. (Submit as .sql or .txt file - Limit insert statements to 25 lines for populating tables)
9. SQL up/down Script to load/migrate in existing data. (Submit as .sql or .txt file - Limit insert statements to 25 lines for populating tables)
10. SQL up/down script of data logic for the external data model. (Submit as .sql or .txt file - Limit insert statements to 25 lines for populating tables)
11. Demonstrate Implementation of the application itself.
12. A team log recording individual and group contributions to the project, including when and by whom.
13. An executive level PowerPoint Presentation.
14. A zip file of all project artifacts, submitted by one team member, for grading.